[INCH-POUND]
A-A-59385
28 September 1999
SUPERSEDING
MIL-G-80179A
11 July 1986

COMMERCIAL ITEM DESCRIPTION

GRINDING MACHINE, CENTER HOLE, VERTICAL, FLOOR-MOUNTED

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-G-80179A for all federal agencies.

1. SCOPE. This commercial item description (CID) covers floor-mounted vertical center hole grinding machines and accessories intended for use in finish grinding workpiece center holes which results in dimensionally accurate reference center holes.

2. CLASSIFICATION. The vertical center hole grinding machines shall be of the sizes shown in table I. The size to be furnished shall be as specified (see 7.2(b)).

TABLE I. Grinding machine sizes.

Size	Center hole diameter (maximum), inches
1	0.5
2	1.5
3	2.0
4	2.5
5	3.0
6	4.0

Note: The range of machine sizes shown in table I imply that the machine offered shall be equal to or larger than the size stated but smaller than the next available size. When the largest machine size is ordered, the machine offered shall not exceed the stated requirements by more than 10 percent.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5678.

AMSC N/A FSC 3415

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3. SALIENT CHARACTERISTICS

3.1 General requirements. The grinding machine shall be new and one of the manufacturer's current models capable of operation in accordance with the requirements herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.2 Components. The grinding machine shall include, as a minimum, the following components:

3.2.1 Base. The base shall support the other elements of the machine and shall incorporate materials, shape, and mass to minimize adverse vibration and deflection. The base shall provide a means for leveling and anchoring the machine.

3.2.2 Column. The column shall be capable of supporting the grinding head assembly, tailstock, and workpiece holding device. Unless otherwise specified (see 7.2(c)), the column length shall be that considered by the manufacturer as standard.

3.2.3 Wheelhead assembly. The wheelhead assembly shall include the following components:

3.2.3.1 Wheelhead spindle. The wheelhead spindle shall be fitted to the wheelhead assembly in a vertical plane and shall have a drive mechanism at one end and a grinding wheel at the other end. The spindle shall be capable of reciprocation along a line at a 30-degree angle to the vertical centerline. The drive motor horsepower and speed shall be that considered by the manufacturer as standard. Automatic mist lubrication shall be provided to the wheelhead.

3.2.3.2 Grinding wheel dresser. The wheelhead assembly shall be fitted with a diamond point grinding wheel dresser capable of dressing the entire working surface of the grinding wheel.

3.2.3.3 Planetary drive. When specified (see 7.2(d)), a grinding wheel planetary drive shall be provided with the drive motor horsepower and rotational speed considered by the manufacturer as standard.

3.2.3.4 Pivoting headstock center. When specified (see 7.2(e)), a pivoting headstock center shall be provided to position noncylindrical workpieces from existing center holes.

3.2.4 Workpiece-holding device. When specified (see 7.2(f)), a workpiece-holding device that is capable of being adjusted both horizontally and vertically shall be provided. The workpiece-holding device shall allow any cylindrical workpiece within the stated machine capacity to be aligned concentrically with the axis of the wheelhead spindle.

3.2.5 Tailstock. The grinding machine shall be fitted with a vertically adjustable tailstock. When specified (see 7.2(g)), the tailstock drive shall be provided with the motor horsepower and tailstock rotational speed considered by the manufacturer as standard.

3.2.6 Dust collector. If required, the grinding machine shall have a dust collector as specified (see 7.2(h)).

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3.2.7 Grinding wheels. If required, grinding wheels shall be supplied with the grinding machine as specified (see 7.2(i)). Unless otherwise specified (see 7.2(j)), grinding wheels shall conform to American National Standards Institute (ANSI) standard B74.2, "Specifications for Shapes and Sizes of Grinding Wheels, and for Shapes, Sizes and Identification of Mounted Wheels", shall be marked in accordance with ANSI standard B74.13, "Markings for Identifying Grinding Wheels and Other Bonded Abrasives", and shall meet the safety requirements of ANSI standard B7.1, "Safety Requirements for the Use, Care and Protection of Abrasive Wheels - Addendum".

3.2.8 Manual controls. Unless otherwise specified (see 7.2(k)), power on/off switches, as a minimum, shall be provided.

3.2.9 Electrical system. Unless otherwise specified (see 7.2(l)), the electrical system shall conform to the ANSI/National Fire Prevention Association (NFPA) standard 79, "Electrical Standard for Industrial Machinery (National Fire Codes, Vol. 3)". The machine shall have dual 230/460-volt, 3-phase, 60-hertz circuit capability and shall be wired for the voltage specified (see 7.2(m)).

3.2.10 Electric motors. The grinding machine drive motor(s) shall have sealed and permanently lubricated ball or roller bearings, shall be energy efficient, and shall conform to the National Electrical Manufacturers Association (NEMA) standard MG-1, "Motors and Generators". Unless otherwise specified (see 7.2(n)), the motors shall have totally enclosed and fan cooled (TEFC) or open dripproof (ODP) enclosures.

3.2.11 Optional accessories. Optional accessories shall be provided as specified (see 7.2(o)).

3.3 Performance. Unless otherwise specified (see 7.2(p)), the grinding machine shall meet the alignment and performance requirements shown in tables II and III, respectively.

TABLE II. Alignment requirements.

Parameter	Maximum allowable deviation
Wheelhead assembly to column alignment	0.0005 inch per 6 inches
Workpiece-holding device to column alignment	0.0008 inch per foot
Axes of wheelhead and tailstock parallel with table movement	0.0001 inch per foot
Tailstock center to planetary head alignment	0.0002 inch TIR[1]
Planetary head square with column	0.0005 inch per 6 inches
Headstock center to planetary head alignment	0.0002 inch TIR

[1] TIR = Total Indicator Reading

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TABLE III. Performance requirements.

Parameter	Maximum allowable deviation
Angle	+0 degrees; –0 degrees, 5 minutes
Concentricity	0.001 inch TIR
Roundness	± 0.00015 inch

3.4 Machine dimensions and weight. If required, the maximum machine dimensions (length, width, and height) and machine weight shall not exceed the restrictions specified (see 7.2(q)).

3.5 Safety and health requirements. The manufacturer shall ensure that the machine and all equipment and accessories used on the machine shall be in compliance with ANSI standard B11.9, "Safety Requirements for the Construction, Care, and Use of Grinding Machines", and Occupational Safety and Health Administration (OSHA) 29 CFR PART 1910, "Occupational Safety and Health Standards for General Industry". If a conflict arises between the ANSI and OSHA standards, the OSHA standards shall apply.

3.6 System of units. The U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices. When only one system of graduation is acceptable, the particular graduation required shall be as specified (see 7.2(r)). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. When specified (see 7.2(s)), measuring and indicating devices shall be graduated in both the US and SI systems of measurement.

3.7 Nameplate. A nameplate shall be attached to the machine. Unless otherwise specified (see 7.2(t)), the nameplate shall contain the following information:

a. Nomenclature
b. Manufacturer’s name
c. Serial number
d. Machine model designation
e. Power input (volts, total amperes, phase, frequency)
f. Short-circuit/over-current rating
g. Contract number or order number
h. National stock number
i. Date of manufacture

3.8 Lubrication plate or chart. When specified (see 7.2(u)), a lubrication plate or chart shall be attached to the machine. The information provided on the plate or chart shall include:

a. Points of lubricant application
b. Servicing interval
c. Type of lubricant(s) with SAE number or lubricant identifier

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4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offerer/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Environmental protection. The item shall meet applicable Environmental Protection Agency (EPA) restrictions in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the user's health and shall be adhered to during the manufacturing, service, transportation, storage, and operation/use of the item.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description; conform to the producer's own drawings, specifications, standards, and quality assurance practices; and be the same product as that which the supplier offers for sale on the commercial market. The government reserves the right to require proof of such conformance.

5.2 Inspection. The grinding machine shall be inspected to determine compliance with all requirements specified in this CID.

5.3 Testing. Unless otherwise specified (see 7.2(v)), the grinding machine shall be tested in accordance with sections 5.3.1 through 5.3.5.

5.3.1 Test setup and conditions. Unless otherwise specified (see 7.2(w)), the manufacturer shall be responsible for supplying all tooling and materials and shall conduct all tests required by the government. The manufacturer shall provide test instruments and equipment with certification traceable to the National Institute of Standards and Technology (NIST) and shall provide certification of calibration as specified (see 7.2(x)). Unless otherwise specified (see 7.2(y)), the grinding machine shall be tested in the environmental conditions at the manufacturer's test facility.

5.3.2 Operational test. The grinding machine shall be operated in accordance with the manufacturer's standard operating procedures. The grinding machine shall be operated at no load for not less than 30 minutes. Proper operation of all controls and features shall be verified during the trial period.

5.3.3. Alignment tests. The grinding machine shall be tested for alignment at the conclusion of the operational test and again at the conclusion of the performance tests. No adjustments of alignment shall be made during or between any of the acceptance tests. The grinding machine shall meet the alignment requirements shown in table II.

5.3.4 Performance tests. Performance tests shall include grinding minimum diameter center holes for the size of machine specified and shall be conducted in accordance with the manufacturer's recommendations and procedures. Following completion of performance testing, the alignment of

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the machine shall be verified again, and the results compared with those recorded after the operational test.

5.3.4.1 Angle test. Using a workpiece located on the tailstock and in the workpiece holding device, finish grind a 60-degree center hole. The center hole shall meet the angle requirements shown in table III.

5.3.4.2 Concentricity test. Using a workpiece 1.0 inch in diameter at one end and 5.0 inches in diameter at the other and locate the workpiece on the tailstock with the workpiece-holding device holding the 5.0-inch-diameter end. Finish grind the center hole. Remove the workpiece, turn it end for end, and with the workpiece-holding device holding the 1.0-inch-diameter end, finish grind the center hole. Check the concentricity of the 1.0 and 5.0 diameters with the workpiece located on centers. The concentricity shall meet the requirements shown in table III.

5.3.4.3 Roundness test. Grind a center hole in the workpiece. Check the roundness of the center hole midway on the cone element. The center hole shall meet the roundness requirements shown in table III.

5.3.5 Optional tests. Optional tests shall be conducted as specified (see 7.2(z)).

5.4 Acceptance. Unless otherwise specified (see 7.2(aa)), preliminary and final acceptance tests shall be conducted at the manufacturer's site. The results of the alignment and performance tests shall be compared with the requirements shown in tables II and III. Failure of the machine to meet the requirements or hold alignment shall be cause for rejection.

6. PACKAGING

6.1 Preservation, packing, and marking. Unless otherwise specified (see 7.2(bb)), preservation, packing, and marking shall be as specified in the contract or order.

7. NOTES

7.1 Sources of documents.

7.1.1 FAR. The FAR may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.

7.1.2 ANSI standards. Copies of ANSI standards may be obtained from the American National Standards Institute, 11 West 42nd Street, New York, NY 10036.

7.1.3 NEMA standards. Copies of NEMA standards may be obtained from the National Electrical Manufacturers Association, 1300 North 17th Street, Suite 1847, Rosslyn, VA 22209.

7.1.4 NFPA standards. Copies of NFPA standards may be obtained from the National Fire Protection Association, 1 Batterymarch Park, Quincy, MA 02269-9101.

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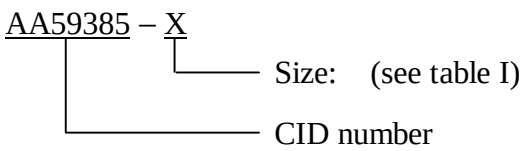
7.1.5 OSHA standards. Copies of OSHA standards may be obtained from the U.S. Department of Labor, 200 Constitution Avenue NW, Room 423, Washington, DC 20210.

7.2 Ordering data. Acquisition documents must specify the following information:

a. Title, number, and date of this document
b. Size of grinding machine required (see 2)
c. Column length, if different (see 3.2.2)
d. Planetary drive, if required (see 3.2.3.3)
e. Pivoting headstock center, if required (see 3.2.3.4)
f. Workpiece-holding device, if required (see 3.2.4)
g. Tailstock drive, if required (see 3.2.5)
h. Dust collector, as specified (see 3.2.6)
i. Grinding wheels, as specified (see 3.2.7)
j. Grinding wheel conformance, marking, and safety requirements, if different (see 3.2.7)
k. Manual controls, if different (see 3.2.8)
l. Electrical system industrial standard, if different (see 3.2.9)
m. Electrical system voltage, as specified (see 3.2.9)
n. Electric motors, if different (see 3.2.10)
o. Optional accessories, as specified (see 3.2.11)
p. Performance, if different (see 3.3)
q. Machine dimensions and weight, as specified (see 3.4)
r. System of units required (see 3.6)
s. Dual system of units, if required (see 3.6)
t. Nameplate, if different (see 3.7)
u. Lubrication plate or chart, if required (see 3.8)
v. Testing, if different (see 5.3)
w. Test responsibility, if different (see 5.3.1)
x. Certification of calibration, as specified (see 5.3.1)
y. Test environmental conditions, if different (see 5.3.1)
z. Optional tests, as specified (see 5.3.5)
aa. Acceptance, if different (see 5.4)
bb. Packaging requirements, if different (see 6.1)

7.3 Part identification number (PIN). The following part identification numbering system is for government purposes and does not constitute a requirement for the contractor.



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MILITARY INTERESTS:

Custodians:
Air Force - 99
Army - AL
Navy - SH

Reviewers:
Air Force - 84
Army - AR, SM
Navy - AS, MC

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 6FET

Preparing activity:
DLA - GS

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